FEDERAL SPECIFICATION

HAMMERS, HAND AND MALLETS, SURFACE PROTECTIVE; HOLDERS AND ADAPTERS, INSERTED HAMMER FACE, AND FACES, HAMMER, INSERTED

This amendment, which forms a part of Federal Specification GGG-H-33a, dated February 10, 1958, was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

Page 2, paragraph 2.1: In the list of *Military Specifications*: Delete MIL-P-15424 in its entirety and substitute:

"MIL-H-15424—Hand Tools, Packaging of".

Page 2, paragraph 3.3, at the end of the paragraph, add the following: "Under visual examination, the hammer shall be well proportioned, that is, the two halves of the hammer shall correspond in size and shape. The longitudinal axis of the head shall be perpendicular to the axis of the handle, and the faces of the hammer shall be in proper alignment with the elliptical or other elongated cross sections of the handle and eye when applicable."

Page 3, paragraph 3.4: At the end of paragraph, add the following: "except type III, class 3 rubber mallets need not be marked."

Page 3, paragraph 3.7 and table I: Delete and substitute the following:

3.7 Rubber for inserts.—Rubber for rubber inserts for hammers shall conform to table I.

TABLE I.—*Physical properties of rubber for rubber inserts*

Tensile strength, p.s.i.	2500 (min.)
Ultimate elongation, percent	200 (min.)
Shore A hardness, instantaneous	95 ± 5
Bashore resilience, 21st impact, percent	15 (max.)
Graves tear strength, pounds per inch thickness	200 (min.)
Cut resistance, pounds per inch, width	2000 (min.)
Impact indentation, inch:	
At 82°F.	0.04 (max.)
At 120°F.	0.04 (max.)
Volume swell after 48 hours in ASTM oil No. 3:	
At 158°F percent	10 (max.)

3.7.1 *Rubber for mallet heads.* Rubber for mallet heads shall conform to table I-A.

TABLE I-A.—*Physical properties of rubber for mallet heads*

Tensile strength, p.s.i.	2500 (min.)
Ultimate elongation, percent	10 (min.)
Shore A hardness, instantaneous	95 ± 5
Bashore resilience, 21st impact, percent	28 (max.)
Volume swell after 48 hours in ASTM oil No. 3:	
At 158°F. percent	12 (max.)
Tear strength, pounds per inch, thickness	200 (min.)

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Page 3, paragraph 3.8.2, lines 4 and 5, parenthetical note: Delete and substitute:

"(Eye numbers do not apply to some tools *with steel or fiberglass handles*)."

Page 3, add as paragraph 3.8.3:

3.8.3 *Tools with fiberglass reinforced plastic handles.* Hammer heads with fiberglass reinforced plastic handles shall have either an eye as specified in 3.8.1 or a hole shall be drilled or reamed approximately three-quarters of the distance through the head in the place where the eye is normally located.

Page 3, paragraph "3.9 Handles", add:

"Each hammer or mallet shall be fitted with a well proportioned wood or fiberglass handle, as specified (see 6.2), or steel handles as specified in 3.9.2."

Page 3, paragraph 3.9.1: Delete the second sentence and substitute the following: "Unless grade AA is specified (see 6.2), the wood handles shall conform to grade A of NN-H-93 for the type and size of head specified."

Page 3, paragraph 3.9.2, add at end of paragraph:

"The grip portion of the handle may be covered with rubber, neoprene or other comparable material."

Page 3, add as paragraph 3.9.3:

3.9.3 *Fiberglass reinforced plastic handles.* Fiberglass reinforced plastic handles shall be well proportioned and shall be molded of a thermosetting type plastic containing a minimum of 60 percent fiberglass. Glass filament fiber orientation shall be unidirectional and longitudinal, and fibers shall be continuous. Fiberglass reinforced plastic handles shall be inserted and bonded by means of a chemical adhesive or otherwise affixed in eyes of tool heads in such a manner that the handle will not loosen under any working condition. There shall be no openings between the handle and head where the eye for the wood handle is normally located. The grip portion shall be fiberglass or other comparable material, bonded to handle. The eye section of the handle may be secured or bonded to the head by means of a chemical adhesive.

Page 4, paragraph 3.10, line 1, delete "3.10 Wedges" and substitute: "3.10 Assembly of wood handles to the head."

Page 4, paragraph 3.10: Add "At the option of the manufacturer, handles may be secured and bonded to the head by means of a chemical adhesive that will prevent the handle from loosening under any normal working condition. Handles assembled with such chemical adhesive shall be subjected to the pull-apart test (see 4.4.2.12)."

Page 4, paragraph 3.11.1.1.1, line 6: After "3.9.1" insert "or fiberglass handle conforming to 3.9.3"

Page 6, paragraph 3.11.2.1.1, line 25: After "3.9.1" insert "or fiberglass handle conforming to 3.9.3."

Page 8, paragraph 3.11.3.1.3, line 14: After "3.9.1" insert "or fiberglass handle conforming to 3.9.3."

Page 10, paragraph 3.12.1, third sentence: Delete and substitute the following: "Both faces shall have a ground or machine finish (see 3.5.1) which shall result in a surface not coarser than that produced by using a 46-grit abrasive or not coarser than 250 micro-inches and the remainder of the head shall have a natural or enamel finish (see 3.5.2 and 3.5.3). Both faces shall be either flat or slightly convex."

Page 10, paragraph 3.12.1, line 9: After "3.9.1" insert "or fiberglass handle conforming to 3.9.3,"

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Pages 12 and 13, paragraph 3.13.1, line 20: After "3.9.1" insert "or fiberglass handle conforming to 3.9.3."

Page 13, paragraph 3.13.2, line 11: After "3.9.1" insert "or fiberglass handle conforming to 3.9.3."

Page 13, table XVI: Delete and substitute new table XVI.

TABLE XVI.—*Type III, class 1 rawhide mallets*

Weight[1] (approx.)	Length of head + 3/8, − 1/4	Diameter of head plus or minus 1/8	Handle length overall + 3 − 1
Ounces	*Inches*	*Inches*	*Inches*
2	2-1/4	1	9
4	2-3/4	1-1/4	10
6	3	1-1/2	10
8	3-1/4	1-3/4	11
11	3-1/2	2	12
22	4-1/2	2-3/4	12
24	4-3/8	2-3/4	13

[1] Weight includes head and handle.

Page 14, paragraph 3.13.3, last sentence: Delete and substitute the following: "The rubber head shall conform to the physical requirements of 3.7.1 and table I-A, and shall withstand the test specified in 4.4.1.9."

Page 14, table XIX: Delete and substitute new table XIX.

TABLE XIX.—*Type III, class 3 rubber mallet*

Weight[1] ± 5	Length of head ± 5/8	Diameter of head + 7/8 − 1/4	Handle length overall + 1 − 2
Ounces	*Inches*	*Inches*	*Inches*
24	4	2	15

[1] Weight includes head and handle.

Pages 14, 15 and 16, paragraphs 4.1 through 4.3: Delete and substitute:

4.1 Inspection responsibility. The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Sampling for inspection and acceptance. Sampling for inspection and acceptance shall be performed in accordance with the provisions set forth in MIL-STD-105, except where otherwise indicated herein.

4.2.1 *Inspection of materials and components.* In accordance with 4.1, the supplier is responsible for insuring that materials and components used were manufactured, tested and inspected in accordance with the requirements of referenced subsidiary specifications and standards to the extent specified herein or, if none, in accordance with this specification.

4.2.2 *Inspection of the end item.* Inspection of the hammers shall be performed as specified in the following paragraphs. A lot shall consist of hammers, mallets, adapters or inserts of the same type, class and style. The sample unit shall be one completely fabricated and assembled hammer, mallet, adapter or insert.

4.2.2.1 *Visual examination.* Each sample unit shall be visually examined for the defects set forth in table XX-A. The inspection level shall be level II, with an acceptable quality level (AQL) of 4.0 defects per hundred units.

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TABLE XX-A.—*Classification of defects*

Examine	Defect
Finish	Not as specified. Ground surfaces, rough, deep scratches, unevenly finished. Coating not smooth, continuous, adherent, blisters, runs, sags, or other detrimental defects to protective value. Cast, molded, extruded, rolled or forged surfaces not free of scale, rust, or other harmful defects. Rubber surfaces, not smooth, pock marks, blisters, fold over lamination, or foreign material embedded in surface.
Material	Not fabricated from specified materials.
Design	Does not conform to type, class, or style specified.
Construction and Workmanship	Hammer not complete, parts missing, improperly fitted or assembled, damaged or non-conforming. Inserts not readily removed, replaced and interchangeable. Evidence of porosity, cracks, burrs, fins, or sharp edges. Threaded studs not securely attached to base of insert, not coated for corrosion resistance, or evidence of non-adherence. Stud threads damaged or incomplete. Threads in head damaged or incomplete. Wood handles not straight grain, not smoothly finished, splits, cracks, or worm holes. Handle not secured and fitted to head as required, wedge or pin missing. Metal handles not knurled. Wedges not flush with front surface of tool eye. End of spiral on rawhide mallets not smooth or securely joined. Voids in face of rawhide mallet. Operation omitted or improperly performed.
Marking	Missing, incorrect or illegible. Plastic inserts not color coded as required.

4.2.2.2 *Dimensional examination.* Examination shall be made for dimensional requirements specified in applicable paragraphs and tables. Any nonconformance shall constitute a defect. The inspection level shall be L4 with an AQL of 4.0 defects per hundred units.

4.2.2.3 *Testing of rubber.* When specified (see 6.2), the manufacturer shall provide a sheet of rubber material 0.07 to 0.08 inch thick, having a total surface area of at least one square foot, together with at least eight solid heads or insert face blanks which shall be of the same composition and cure, and prepared at the same time from material used in the heads or inserts furnished. Each sample rubber insert shall be tested as specified in 4.4.1 through 4.4.1.8 to verify compliance with the physical requirements specified in 3.7 and table I. Each sample rubber mallet shall be tested as specified in 4.4.1.9 through 4.4.1.9.6 to verify compliance with the physical properties specified in 3.7.1 and table I-A.

4.2.2.4 *Testing of plastic.* When specified (see 6.2), the manufacturer shall provide at least eight solid heads or insert face blanks (specimens) of the plastic material which shall be of the same composition and cure and prepared at the same time as the material used in the heads or inserts furnished. Each sample shall be tested as specified under 4.4.2 to verify compliance with this specification.

4.2.2.5 *Pull-apart and striking test.* Testing shall be performed on each sample hammer (when applicable) for determination of requirements as specified in 4.4.2.6 and 4.4.2.12. The inspection level shall be I.4 with an AQL of 2.5 defects per hundred units.

4.3 **Preparation for delivery.** Inspection to determine compliance with requirements for preparation of delivery shall be performed in accordance with Specification MIL-H-15424.

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Page 16, paragraph 4.4.1, lines 1 and 2, in two places, delete the following: "mallet heads and,".

Page 17: Following paragraph 4.4.1.8, insert the following new paragraphs:

4.4.1.9 *Tests for rubber mallet heads.* Rubber for rubber mallet heads shall be tested in accordance with 4.4.1.9 through 4.4.1.9.6 to verify compliance with 3.7.1 and table I-A.

4.4.1.9.1 Tensile strength and ultimate elongation shall be determined in accordance with 4.4.1.1.

4.4.1.9.2 Hardness measurements shall be made in accordance with 4.4.1.2.

4.4.1.9.3 Resilience test shall be made in accordance with 4.4.1.3 and 4.4.1.3.1.

4.4.1.9.4 Determination of tear resistance shall be made in accordance with 4.4.1.4 through 4.4.1.4.3 except that ASTM Tear Die "C" may be used in lieu of the Graves Angle Die specified in 4.4.1.4.1.

4.4.1.9.5 The swelling test shall be made in accordance with 4.4.1.5 through 4.4.1.5.2.1.

4.4.1.9.6 The rubber mallet heads shall be in accordance with 4.4.1.8.

Page 18, paragraph 4.4.2.6, line 3: After the word "blows" insert "by continuous hand striking or the mechanical equivalent". Delete the second sentence beginning with "The blows" and ending with "striking."

Page 19, add new paragraph 4.4.2.12.

4.4.2.12 *Pull-apart test.* Each hammer selected in accordance with 4.2.4 that has the handle assembled with a chemical adhesive bond shall have the head and handle securely held in a standard tensile-testing machine, or some other method may be employed. Hammers with heads weighing over 12 ounces, nominal weight, shall not pull apart or loosen when subjected to a pulling pressure of 2,250 pounds. Hammers with heads weighing 12 ounces or under, nominal weight, shall not pull apart or loosen when subjected to a pulling pressure of 750 pounds.

Page 19, paragraphs 5.1.1, 5.2.1, 5.2.2, and 5.3.1: Delete "MIL-P-15424" and substitute "MIL-H-15424".

Page 20, paragraph 6.2, option (*c*): Delete and substitute the following:

(*c*) For hammers or mallets with wood handles, if grade AA handles are required, so state (see 3.9.1).

Page 20, paragraph 6.2: Add new items (*k*), (*l*) and (*m*) as follows:

(*k*) For types I and III hammers, specify either wood handles (see 3.9.1) or fiberglass handles (see 3.9.3).

(*l*) If a sheet of rubber material is required for testing (see 4.2.2.3).

(*m*) If 8 sample plastic heads or inserts are required for testing (see 4.2.2.4).

Page 20: Insert the following paragraph:

6.4 Transportation description. Transportation descriptions and minimum weights applicable to this commodity are:

Types I and II:
Rail:
Hammers, hand, not otherwise indexed by name.
Carload minimum weight 30,000 pounds.

Motor:
Hammers, hand, not otherwise indexed.
Truckload minimum weight 30,000 pounds, subject to Rule 115, National Motor Freight Classification.

Types III and IV:
Rail:
Tools, not otherwise indexed by name.
Carload minimum weight 30,000 pounds.

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Motor:

Tools, hand, not otherwise indexed.
Truckload minimum weight 30,000 pounds, subject to Rule 115, National Motor Freight Classification.

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